Org Knowledge Credibility — Presentation & Demo Script

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Before you start (15 minutes before)

The talk runs about 15 minutes: roughly 7 minutes of explanation and 8 minutes of live demo. Get the stack warm before anyone arrives, because the first NLI call loads a 0.7 GB model.

- [ ] Start everything: `docker compose up -d --build` from the project folder
- [ ] Check the services are up: Streamlit UI on http://localhost:8502, API docs on http://localhost:8001/docs, Elasticsearch on port 9200, PostgreSQL on port 5433
- [ ] Confirm the corpus is indexed: the top bar in the UI should show a healthy backend and a document count. If it is empty, run `docker compose exec backend python -m app.cli ingest --full`
- [ ] Run one claim end to end so the NLI model is downloaded and loaded into memory
- [ ] Copy the demo claims (Live demo section) into a notes file so you can paste them, not type them
- [ ] Optional: start Ollama with `gemma4:e4b` only if you plan to show the Ask tab or deep review. It takes about 40 s per passage on CPU, so do not show it live unless it is pre-run
- [ ] Close other heavy apps; Docker needs at least 8 GB of memory

Fallback: if the stack fails on the day, use the Evaluation results and screenshots of a finished assessment. Every assessment is stored, so you can also open a previous one instead of running a new one.

1. Opening — the problem (1 minute)

Say:

> "Every company has rules written down somewhere. The trouble is, they are written down in many places, and those places disagree.
>
> Take a simple question: *do discounts above 15% need Finance approval?* The current policy says yes. An older policy, still sitting on the drive, says the threshold is 10%. A Slack thread says the Head of Sales approves urgent deals herself and sorts the paperwork out later. And the CRM records show what actually happened.
>
> So which one do you believe? A search engine would just show you whichever one matches your words best. That's not good enough.
>
> We built a platform that takes a claim like that and tells you whether it's credible: **supported, contradicted, contested, or not enough evidence**. It also shows you exactly which passages it used, how much each one counted, and where a person still needs to decide."

Show: the Check a claim tab with the discount claim already in the box. Don't run it yet.

2. The core idea — relevance is not credibility (1 minute)

Say:

> "The one idea to remember: **a passage being relevant doesn't make it credible.**
>
> A superseded policy can match your question word for word, and it's still the wrong answer. Ten Slack messages repeating the same rumour are one source, not ten. A draft that was never approved is a proposal, not a rule.
>
> So we split the job in two. Search finds candidate evidence. A separate reasoning engine then judges every passage: does it support the claim, contradict it, or neither? How much authority does this source have *for this kind of claim*? Is it still in force? Is it just a copy of something we've already counted?"

Three things to stress (these are what make it different from a search tool or a chatbot):

1. **Authority depends on the question.** A policy is the best evidence of what the rule *is*. A CRM record or an interview is the best evidence of what people actually *do*.
2. **Time matters.** Superseded, stale, draft and undated documents count for less. Evidence dated after the question's as-of date is excluded.
3. **Copies don't add up.** Sources that come from the same origin are merged into one source lineage, so repetition can't inflate the score.

Optional line: "Everything runs locally. No document leaves the machine, and no external AI service is needed."

3. How it works (2 minutes)



claim assessment pipeline · 6 steps + audit

Put this picture on screen and walk left to right, top row then bottom row.

Say:

> "**One, ingest.** We read 45 files: policies, procedures, meeting notes, interviews, Slack and email, CRM deals, tasks and an audit log. Documents are split at their headings, never mid-sentence, and each piece keeps its date, owner and status, like draft or superseded.
>
> **Two, relate.** We build a map of how documents link: this policy supersedes that one, this procedure implements that policy, this Slack thread follows up that meeting. A newer date on its own never means something is superseded.
>
> **Three, retrieve.** For a claim we run a normal search, plus a deliberate search for counter-evidence, and we pull in linked documents.
>
> **Four, reason.** Each passage is labelled: supports, contradicts, neutral or ambiguous.
>
> **Five, weigh.** Each passage gets a weight based on its source, its date and how direct it is. Copies are merged.
>
> **Six, decide.** We combine it into a score, a verdict and a list of who should check what.
>
> Underneath it all, every assessment is saved with its evidence and settings, so any answer can be re-run and checked later."

Optional extra: there's also a claim-suggestion feature that proposes claims to test from policies and meeting decisions. Suggestions are never treated as facts; a person chooses which to assess.

4. How the score works (2 minutes)

Say:

> "The score isn't a black box. It's three simple steps, and every number lives in one config file.
>
> **First, every passage gets a label.** Supports, contradicts, neutral or ambiguous, plus a strength. Rules look for things like a matching threshold, the same approver, negation, or workaround language like 'after the fact'. An NLI model, which judges whether one sentence entails or contradicts another, can step in where the rules are unsure.
>
> **Second, every passage gets a weight.** We multiply five things: how much authority the source has for this kind of claim, whether it's still current, how direct it is, how complete its metadata is, and how strong the label was.
>
> **Third, we add it up.** Copies from the same origin are merged so they count roughly once. Then all the support goes on one side and all the contradiction on the other."

Put the formula on screen:

```
\text{weight} = \text{authority} \times \text{temporal} \times
\text{directness} \times \text{provenance} \times \text{strength}
```

```
\text{score} = \frac{S + 0.25}{S + C + 0.5}
```

> "S is the total supporting weight and C the total contradicting weight. With no evidence at all the score sits at 0.5, and it moves towards 1 or 0 only as evidence builds up. We also report a range around it. A wide range means we found little evidence, so the answer is shaky."

Check, in this order	Verdict
Too little evidence weight (S + C < 0.3), or the evidence covers less than 30% of what the claim says	Not enough evidence

Check, in this order	Verdict
Score 0.7 or higher	Supported ("Likely" if only weak sources or partial coverage)
Score 0.3 or lower	Contradicted
Strong weight on both sides	Contested, flagged for human review

> "One honest caveat: this score is a **weighted summary of evidence, not a probability** that the claim is true. We say that in the interface too."

If someone asks why a policy counts 1.0 but a Slack message 0.3: those are judgement calls set in `config/credibility.yaml`. We tested how much the verdicts move when you change them (sensitivity analysis in the evaluation).

5. Live demo (8 minutes)

Four claims, each showing one behaviour. The verdicts and scores below come from a live run on 21 Oct 2026 (rules + NLI, hybrid search, 15 pieces of evidence). Each claim takes 6–20 seconds. Use the example pills or paste the text exactly.

#	Claim (paste exactly)	Live result	What it shows
A	Head of Finance and People Ops must approve new-business discounts of 15 percent or more in the CRM before Closed Won.	Supported, score 0.89	Supersession: the old policy is set aside
B	On genuinely urgent deals, Head of Sales may approve discounts of 15 percent or more outside the CRM, with paperwork later.	Supported, score 0.86, flagged for review	Practice vs policy, copies merged
C	All renewal discounts stay within the 10 percent cap.	Contested, score 0.57, human review	Real conflict the system won't hide
D	Northfield offers new customers a 30-day free trial.	Not enough evidence	A high score is not enough on its own

Demo A — the rule, and the old rule it replaced (2 min)

Do: Check a claim → click the **Discount approval** example → **Check credibility**.

Say while it runs: "It's searching by keyword and by meaning, plus a deliberate search for counter-evidence: words like exception, workaround, superseded. That stops it only finding passages that agree."

Point at: the verdict card (Supported, 0.89), then the supporting and against columns.

> "The current policy, POL-01, supports it, and so do CRM deal records and an audit log entry. One passage goes against it: POL-02, the old policy with the 10% threshold. The system knows POL-01 explicitly supersedes POL-02, so the old one is treated as history, not as a current contradiction. That's why the score stays high."

Open: Full evidence trail → show a passage's label, scope and the reason it was given.

Demo B — what people actually do (2 min)

Do: click the **Urgent-deal workaround** example → **Check credibility**.

> "This is a different kind of claim. It's not about the written rule, it's about practice. So now interviews, Slack threads and deal records count for more, and the policy counts for less.
>
> It comes back Supported, from the interview, the Slack thread and deal DEAL-014. But look at this note: the Slack thread COMMS-02 and the meeting MTG-02 are linked, so they're **counted as one source lineage**, not two. And it's flagged for review because the evidence range is wide."

Point at: the merged copies note and the "A person should decide" notice.

Demo C — a genuine conflict (2 min)

Do: click the **Renewal discount cap** example → **Check credibility**.

> "Here the documents really disagree. The renewal policy POL-06 sets a 10% cap, and it's also past its review date. But deal records and a Slack thread show renewals above 10%, and a meeting, MTG-10, discussed it.
>
> Nothing records which one governs: no supersession, no approval, no staleness flag. So rather than guess, the system says **Contested**, names the two documents, and tells you to ask their owners which one is in force. That's the behaviour we want: surface the conflict, don't paper over it."

Demo D — when the score looks good but the answer is 'we don't know' (1 min)

Do: paste *Northfield offers new customers a 30-day free trial.* → **Check credibility**.

> "The raw score is actually high, 0.87, because some audit records mention a 30-day period. But none of the decisive evidence mentions a free trial at all; it covers under 30% of the claim. So the verdict is **Not enough evidence**. Absence of evidence is never reported as a contradiction."

Wrap the demo (1 min)

- **Evidence tab:** "These are the 45 source files: policies, procedures, meetings, interviews, Slack and email, CRM, tasks and the audit log. Every one keeps its date, owner and status." Show a superseded or draft file.
- **"Was this right?" buttons** under a verdict: "Users can correct a verdict. Those labels feed the calibration check."
- **Optional, only if pre-run:** the Ask tab, where the local LLM (gemma4) rewrites a free-form question into a testable claim. It takes about 40 s per passage on CPU, so avoid doing it live.

Avoid live: *Feature requests are currently tracked on a Trello board.* It currently comes back Supported (0.85) though the reference answer is Contradicted, because a stale procedure (PROC-03) still describes Trello. It's a good limitation example, but only if you introduce it as one.

How it works tab: it now matches this script (rules + NLI model, superseded files ×0.2), so you can use it as a backup slide for section 3.

6. Does it work? Results and limits (2 minutes)

Say:

> "We tested it on 52 labelled claims, split into a development set we tuned on and a held-out test set we only scored once, at the end."

Result	Rules only	Rules + NLI (default)
Verdict accuracy, held-out test (4 classes)	52%	57% (95% CI 39–74%)
Passage stance accuracy, test	79%	89%
AUC of the score for "supported"	0.81	0.86

Result	Rules only	Rules + NLI (default)
Brier score (lower is better; 0.25 = guessing the base rate)	0.214	0.145

> "Against the two earlier versions of this project on the same 54 claims, it gets all 4 client-labelled claims right, and 76% on the simpler credible-or-not decision, against 65% and 50% for the earlier versions. The biggest single improvement came from adding the NLI model: the rules alone trusted word overlap too much.
>
> We also tried training a logistic regression on top. With only about 30 training claims it did worse than the transparent formula, so we kept the formula."

Be upfront about the limits (pick two or three):

- Only 4 claims have client-supplied labels; the other 48 were written by the project author and need independent review.
- The corpus is small, about 45 files, and synthetic. Results may not transfer to a real company archive.
- With claims of mixed evidence, it often says Contested when the reference answer is Supported or Contradicted.
- The rules rely on wording; they miss paraphrase and implicit negation. NLI helps but isn't perfect.
- It's single-user with no login, and the local LLM is slow on CPU (about 40 s per passage).

7. Close and questions (1 minute + Q&A)

Say:

> "To sum up: search finds the evidence, but credibility needs judgement. Who wrote it, is it still in force, is it a rule or a practice, is it just a copy of something else? We made every one of those judgements explicit, adjustable and auditable. Every assessment stores its evidence, labels, weights and config, so you can reproduce any answer later. And where the documents genuinely disagree, it tells you who to ask instead of guessing."

Likely questions:

Question	Suggested answer
Why not just ask ChatGPT?	An LLM gives a fluent answer but can't show why one source outranks another, and the data must stay local. Here every weight is visible and reproducible. The LLM is optional and never sets the score.
Is the score a probability?	No. It's a weighted summary of the evidence. The range around it shows how much evidence there was.
Where do the weights come from?	They're stated assumptions in `config/credibility.yaml`. We ran a sensitivity analysis to see how much verdicts change when they move.
How does it know a policy is superseded?	From metadata, folder and file names, and explicit "supersedes" links. A newer date alone never counts as supersession.
What if two sources say the same thing?	If they share an origin (a follow-up, near-identical wording, the same person), they count as one source lineage.
Can it handle a new company's documents?	Yes. Point `DATA_DIR` at the folder and run a full ingest. Folder rules, aliases and roles are set in `config/`.
What happens when it's wrong?	Users mark it with "Was this right?". Those labels feed the calibration check, and the config can be retuned on the dev set.